

Dimensional properties of Laguerre–Gaussian vortex beams

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We derive and analyze the dimensional properties of Laguerre–Gaussian vortex beams theoretically and numerically. The analytical expressions of inner and outer radii are given out and proved to be proportional to the transverse beam size, when the topological charge remains. The ratio of the outer and inner radius only depends on the topological charge, having nothing to do with the waist radius and the propagation distance, and the ratio decreases as the topological charge increases. Using a spatial light modulator loaded with computer-generated holograms, we generate vortex beams. The experimental results are in good agreement with our numerical simulations. This research may provide useful insights into the study of the optical vortex beam and its further applications. © 2017 Optical Society of America

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1. INTRODUCTION

Optical vortices possess a special helical phase structure and on-axis phase singularity where the intensity is zero and phase value is undefined [1,2]. For such beams, the orbital angular momentum per photon is $m\hbar$, which corresponds to azimuthal phase profile $\exp(im\theta)$ [3,4], where m is the optical vortex topological charge. This indicates m times that phase should change by 2π in winding around the optical beam axis [5].

With these unique and novel properties, optical vortices have been widely studied and applied in recent years, especially in the fields of trapping and manipulation of microscopic particles, which is known as optical tweezer or optical wrench [6–8]. In addition, such beams have also shown broad application prospects in optical communication systems through free space and fiber [9–11]. Good alignment and dimension matching [12,13], which requires a comprehensive understanding of the intensity distribution and dimension properties within optical vortices, are important to well couple the vortex beams into the fiber, which has an annular shape.

Phillips and Andrews derived the expression of the effective radius for all orders of Laguerre–Gaussian (LG) beams [14]. In the LG mode, the radius of the beam, defined as the distance from the maximum intensity to the center, was given out in Ref. [2]. For such beams, the intensity profile and beam divergence are strongly dependent on their topological charges [15]. Reddy *et al.* studied the spatial profile of vortices embedded in a Gaussian host beam using two novel and measurable parameters (inner and outer radii) [16], and further studied the change

rate of the inner and outer radii, divergence, and their variation with the order [17]. In addition, the dependence of the dark region and bright ring areas on the different effective orders of coherent vortices was studied by pixels [18]. However, there is not a more thorough study on the ratio of inner and outer radii of Laguerre–Gaussian vortex beams that may be used to improve coupling of the orbital angular momentum beams to fibers with a desired annular index profile.

In this paper, the dimensional properties of the general LG vortex beams are proposed. We derive the analytic expressions of the inner and outer radii. When the topological charge remains constant, the inner radius is directly proportional to the transverse beam size, as is the outer radius, and the ratio of the outer and inner radius is a function of the topological charge only, no matter what the waist radius and propagation distance are. Using a spatial light modulator (SLM) loaded with computer-generated holograms (CGHs), we generate optical vortices whose ratios are in a good agreement with the numerical simulations.

2. THEORETICAL ANALYSIS

As the most typical representative of the optical vortices, the LG vortex beam has been widely used in the laboratory owing to its circular symmetry as well as its ready availability. It can be generated by a geometric model conversion method [19], spiral phase plates [20,21], and computer-generated holograms [22,23]. A typical LG vortex beam has p (with $p + 1$ concentric rings) radial nodes in its amplitude profile. Here, it is limited to

having single-ring mode with $p = 0$. The initial scalar electric amplitude of a LG vortex beam with topological charge m can be expressed in cylindrical coordinates as

$$\text{LG}_0^m(r, \theta, z) = \left(\frac{2}{\pi|m|!}\right)^{\frac{1}{2}} \cdot \frac{1}{w(z)} \left(\frac{r\sqrt{2}}{w(z)}\right)^{|m|} \exp\left(\frac{-r^2}{w(z)^2}\right) \cdot \exp(-im\theta) \exp(i[kz - kr^2/(2R)]) \exp(i\varphi), \quad (1)$$

where

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2}, \quad R_s = z[1 + (z/z_R)^2],$$

$$z_R = \frac{\pi w_0^2}{\lambda}, \quad \varphi = (|m| + 1)\tan^{-1}(z/z_R). \quad (2)$$

In Eqs. (1) and (2), r , θ , and z represent the cylindrical coordinate parameters, $w(z)$ denotes the transverse beam size with w_0 being the waist radius at $z = 0$, k is the wave number, Z_R represents the Rayleigh range of the Gaussian envelope, R_s is the curvature radius of the wavefront, and φ denotes the Gouy phase, which changes with the propagation distance from the waist position. Accordingly, its intensity is

$$I(r, \theta, z) = \left(\frac{2}{\pi|m|!}\right) \cdot \frac{1}{w(z)^2} \left(\frac{r\sqrt{2}}{w(z)}\right)^{2|m|} \exp\left(-\frac{2r^2}{w(z)^2}\right). \quad (3)$$

The radius of the LG vortex beam, defined by the distance from the beam center to the points where the maximum intensity is found, is given by

$$r_0 = \sqrt{\frac{|m|}{2}} \cdot w(z). \quad (4)$$

The inner and outer radii (r_1 , r_2) are the radial distances at which the intensity falls to $1/e^2$ of the maximum intensity at r_0 . Here, r_1 is the point closer to the center and r_2 is the point farther from the center. Then their light intensity, $I(r_i, \theta, z)$ ($i = 1, 2$), satisfies the relations

$$I(r_i, \theta, z) = \frac{1}{e^2} I(r_0, \theta, z). \quad (5)$$

Taking the LG vortex beam LG_0^4 as an example, the light intensity distribution of an optical vortex and the radial parameters have been shown in Fig. 1.

Setting m positive for simplicity, the insertion of Eqs. (3) and (4) into Eq. (5) gives

$$\left(\frac{r_i}{w(z)}\right)^{2m} \cdot \exp\left(\frac{-2r_i^2}{w(z)^2}\right) = \left(\frac{m}{2}\right)^m \exp(-m-2). \quad (6)$$

For convenience, we define $t = r/w(z)$ ($0 < t < \infty$); then

$$t^{2m} \cdot \exp(-2t^2) = \left(\frac{m}{2}\right)^m \exp(-m-2). \quad (7)$$

Here, let a continuous function

$$f(t) = t^{2m} \exp(-2t^2). \quad (8)$$

It is found that $f(t)$ monotonically increases in the interval $(0, \sqrt{m/2})$ and monotonically decreases in the interval $(\sqrt{m/2}, \infty)$. Its maximum gives

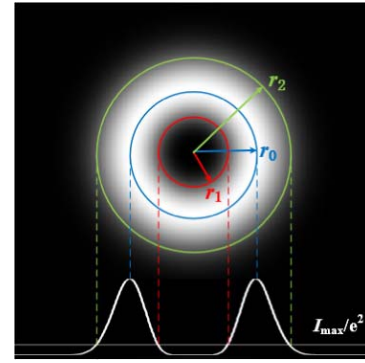


Fig. 1. Intensity distribution and radial parameters of single-ring LG mode with topological charge 4.

$$f\left(\sqrt{\frac{m}{2}}\right) = \left(\frac{m}{2}\right)^m \cdot \exp(-m), \quad (9)$$

and the limiting values of the function $f(t)$ are

$$\lim_{t \rightarrow 0} f(t) = 0, \quad \lim_{t \rightarrow \infty} f(t) = 0. \quad (10)$$

Here, $f(t)$ starts at 0 for 0 t , rises to a maximum, and then decays back to 0 for large t , so that the Eq. (7) is fulfilled at two points: t_1 denotes the smaller solution and t_2 denotes the larger one, which can be solved analytically in terms of the Lambert W function, and the solutions for m return

$$t_1 = \sqrt{\frac{m}{2}} \cdot \sqrt{-W\left(0, -\frac{(e^{-m-2}m^m)^{1/m}}{m}\right)},$$

$$t_2 = \sqrt{\frac{m}{2}} \cdot \sqrt{-W\left(-1, -\frac{(e^{-m-2}m^m)^{1/m}}{m}\right)}, \quad (11)$$

so that the analytical expression of the inner and outer radii

$$r_1 = \sqrt{\frac{m}{2}} \cdot \sqrt{-W\left(0, -\frac{(e^{-m-2}m^m)^{1/m}}{m}\right)} \cdot w(z),$$

$$r_2 = \sqrt{\frac{m}{2}} \cdot \sqrt{-W\left(-1, -\frac{(e^{-m-2}m^m)^{1/m}}{m}\right)} \cdot w(z). \quad (12)$$

In the function $W(k, \arg)$, k is an integer and has infinitely many branches, but the two real solutions needed here always involve $k = 0$ and $k = -1$.

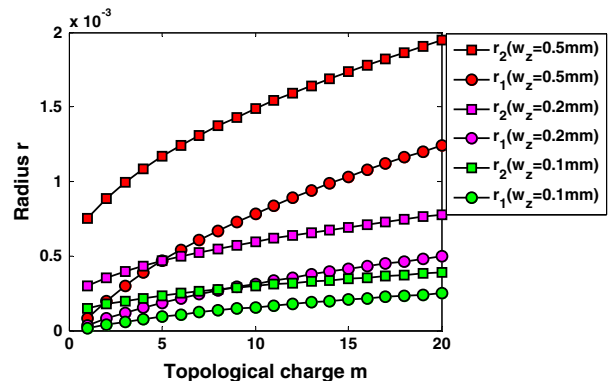


Fig. 2. Variations of the inner and outer radii of LG vortex beams with topological charge.

According to Eq. (12), when transverse beam size $w(z)$ is set at 0.1, 0.2, and 0.5 mm respectively, the inner and outer radii for LG vortex beams with topological charge $m = 1 - 20$ are calculated and plotted in Fig. 2. We can see the inner and outer radii increase with the increase of the topological charge.

As we know, for LG vortex beams with a certain topological charge m , the inner and outer radii $r_{1,2}$ vary with waist radius w_0 and propagation distance z . However, according to Eq. (11), $r_1/w(z)$, $r_2/w(z)$, and r_2/r_1 are fixed values.

3. NUMERICAL SIMULATION

Considering the LG modes with a certain topological charge, we use $m = 4$ as an example. According to Eqs. (2) and (6), by setting different values of waist radius w_0 and propagation distance z , the numerical values of $w(z)$, r_1 , and r_2 can be calculated, as shown in Tables 1 and 2. These values are the same as the values calculated from Eqs. (11) and (12).

In Table 1, we keep the propagation distance $z = 0.4$ m, and the waist radius parameter w_0 is set as 0.3, 0.35, 0.4, and 0.45 mm,

Table 1. Numerical Values for LG Mode with $m = 4$, $z = 0.4$ m, and $w_0 = 0.3, 0.35, 0.4$, and 0.45 mm, Respectively

w_0 (mm)	$w(z)$ (μm)	r_1 (μm)	r_2 (μm)	$r_1/w(z)$	$r_2/w(z)$	r_2/r_1
0.30	402.43	312.61	873.86	0.7768	2.1715	2.7954
0.35	418.79	325.29	909.33			
0.40	447.74	347.80	972.26			
0.45	484.23	376.15	1051.49			

Table 2. Numerical Values for LG Mode with $m = 4$, $w_0 = 0.4$ m, and $z = 0.3, 0.4, 0.5$, and 0.6 mm, Respectively

z	$w(z)$ (μm)	r_1 (μm)	r_2 (μm)	$r_1/w(z)$	$r_2/w(z)$	r_2/r_1
0	300.00	233.04	651.45	0.7768	2.1715	2.7954
0.3	361.21	280.59	784.36			
0.4	402.43	312.61	873.86			
0.5	449.91	349.49	976.97			
0.6	501.88	389.86	1089.82			

and 0.45 mm. In Table 2, we keep the waist radius w_0 at 0.3 mm and the propagation distance z is set as 0, 0.3, 0.4, 0.5, and 0.6 m, respectively. Here, the wavelength λ is supposed to be 632.8 nm.

Tables 1 and 2 show that, for an LG vortex beam with certain topological charge, the inner radius r_1 and outer radius r_2 increase with the increase of the waist radius w_0 and propagation distance z . However, the ratios of $r_1/w(z)$, $r_2/w(z)$, and r_2/r_1 are unchanged.

Figure 3 shows the variation of the inner and outer radii at the transverse beam size $w(z)$ for LG vortex beams with $m = 4$ numerically. In Fig. 3(a), the propagation distance z is 0.4 m, and the waist radius parameter w_0 changes from 0.1 to 2 mm with an interval of 0.1 mm. Accordingly, the variation of $w(z)$ ranges from 0.35 to 2.1 mm. In Fig. 3(b), the waist radius w_0 is 0.3 mm, and distance ranges from 0 to 1.9 m with an interval of 0.1 m. Accordingly, the variation of $w(z)$ ranges from 0.25 to 1.35 mm. It can be observed that, as the waist radius or propagation distance increases, the inner and outer radii always increase linearly with the transverse beam size. The lines pass through the origin and their slopes are 0.7768 and 2.1715, respectively. The ratios of the outer and inner radius are all 2.7954. These numerical results have good agreement with

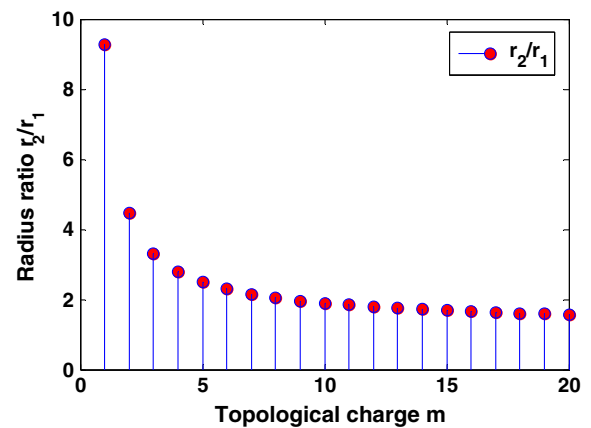


Fig. 4. Relation between radius ratio and topological charge.

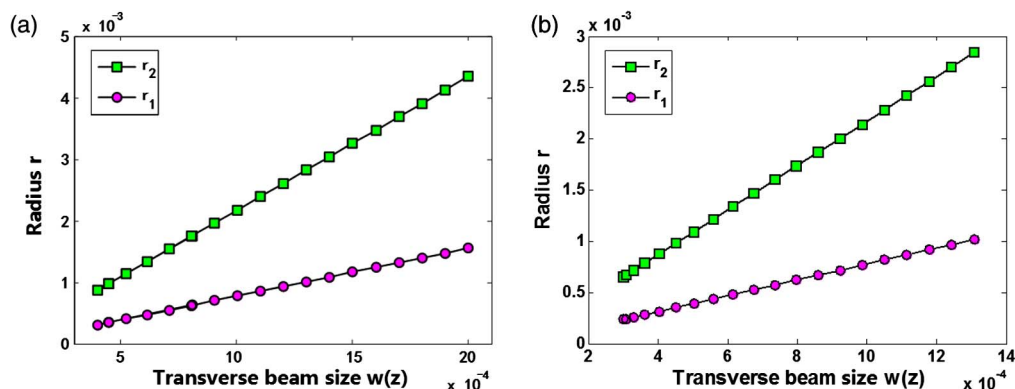


Fig. 3. Variations of the inner and outer radii of LG beams for order 4 with transverse beam size $w(z)$ for different values: (a) propagation distance z and (b) waist radius w_0 .

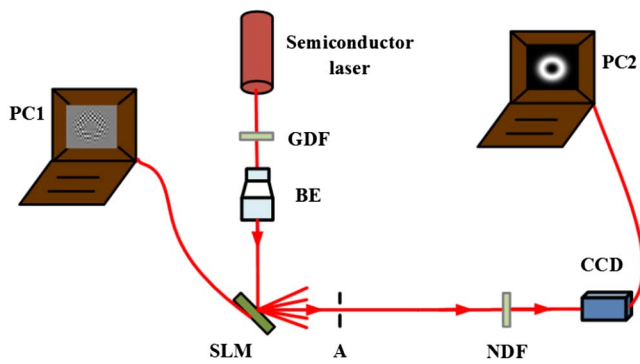


Fig. 5. Schematic of experiment setup. PC1, PC2, computer; GDF, gradient density filter; BE, beam expander system; NDF, neutral density filter; SLM, spatial light modulator; CCD, charge-coupled device.

the theoretical proof that $r_1/w(z)$ and $r_2/w(z)$ are fixed values for LG vortex beams with a certain topological charge m .

The relation between the radius ratio r_2/r_1 and topological charge m is shown in Fig. 4. The numerical values of the radius ratio for the LG vortex beams with topological charge $m = 1 - 20$ are calculated. As shown by the changing trend, the radius ratio decreases and asymptotically approaches a constant with the increase of the topological charge number. This is shown in the intensity distribution of LG vortex beams, as the light ring area decreases while the dark area increases.

4. EXPERIMENTAL SETUP AND RESULTS

A. Experimental Setup

The schematic of the experimental setup is shown in Fig. 5. The light emitted from the source He-Ne laser is attenuated appropriately by a gradient density filter and then expanded by the beam expander. The expanded beam propagates towards an electronically addressed reflection-type SLM (Holoeye, PLUTO, pixel number 1920×1080) controlled by a computer PC1, where the holograms of the LG vortex beams are loaded. The holograms are generated by a CGH method based on conjugate-symmetric extension [23]. The generated LG

vortex beams are selected by an aperture. In the experimental measurement, the LG vortex beams are recorded by a CCD camera (Pike 505B), which is controlled by a computer PC2. A neutral density filter is placed in front of the CCD to avoid saturation of the CCD.

B. Experimental Results and Analysis

With the CGHs of LG vortex beams of $m = 4$ and different waist radii loaded to the SLM, we generate the optical vortices and acquire the images at $z = 0.4$ m by CCD. Figures 6(a)–6(d) are the intensity patterns at waist radii 0.3, 0.35, 0.4, and 0.45 mm, respectively. Figure 6(e) shows the normalized transverse intensity profiles.

From the line profiles in Fig. 6(e) and the inner and outer radii, the experimental ratio can be obtained from the pixel number, as shown in Table 3.

With the CGH of an LG vortex beam with $m = 4$ and $w_0 = 0.3$ mm loaded to the SLM, we generate the optical vortex and acquire the images at different propagation distances from the CCD. Figures 7(a)–7(d) show the intensity pattern at propagation distances 0.3, 0.4, 0.5, and 0.6 m, respectively. It can be seen that the vortex beams expand in size with the increase of propagation distance due to light diffraction. Figure 7(e) shows the normalized transverse intensity profiles.

From the line profiles in Fig. 7(e) and the inner and outer radii, the experimental ratio can be obtained from the pixel number, as shown in Table 4.

From Tables 3 and 4, all the experimental ratios approach the numerical value—namely, the ratio of the outer and inner radius is constant, which is unrelated to the waist radius and propagation distance, when the topological charge remains.

With the CGHs of LG vortex beams with $w_0 = 0.3$ mm and different topological charges loaded to the SLM, we

Table 3. Result for the Generated LG Vortex Beams with $m = 4$, $z = 0.4$ m, and Different Waist Radii

w_0	r_1	r_2	r_2/r_1 (expt)	r_2/r_1 (theo)	$\Delta(r_2/r_1)$
0.30	92.5	256.5	2.7730	2.7954	-0.0224
0.35	98.5	273.0	2.7716		-0.0238
0.40	111.5	315.0	2.8251		0.0297
0.45	119.0	333.5	2.8025		0.0071

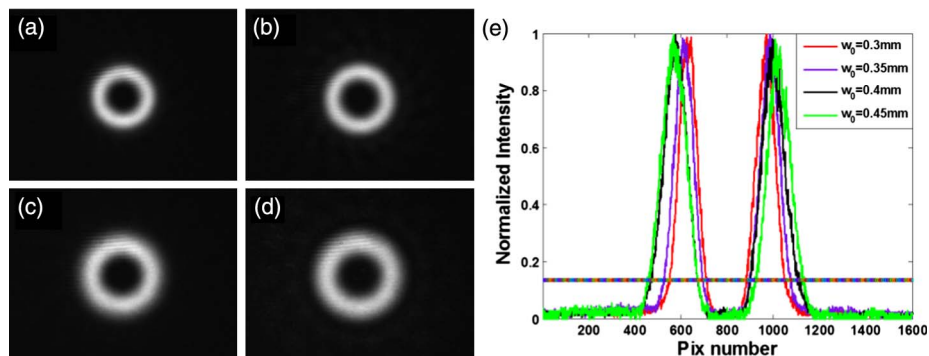


Fig. 6. Generated LG vortex beams with $m = 4$ and different waist radii: (a) $w_0 = 0.3$ mm, (b) $w_0 = 0.35$ mm, (c) $w_0 = 0.4$ mm, (d) $w_0 = 0.45$ mm at $z = 0.4$ m, and (e) the normalized transverse intensity profiles.

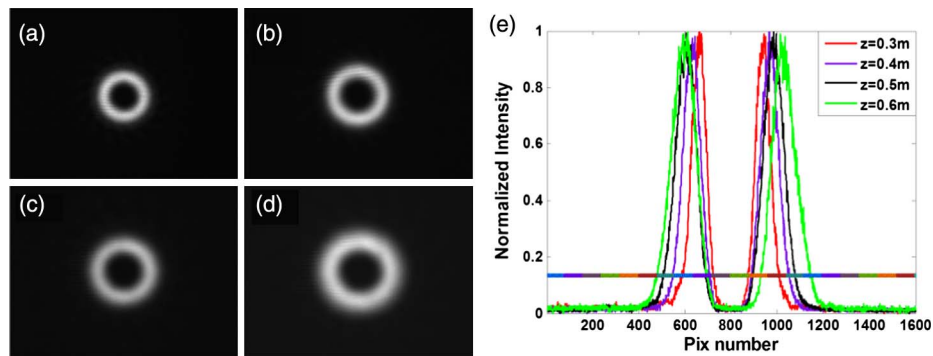


Fig. 7. Generated LG vortex beams with $m = 4$ and $w_0 = 0.3$ mm at different propagation distances: (a) $z = 0.3$ m, (b) $z = 0.4$ m, (c) $z = 0.5$ m, (d) $z = 0.6$ m, and (e) the normalized transverse intensity profiles.

Table 4. Result for the Generated LG Vortex Beams with $m = 4$, $w_0 = 0.3$ mm, and Different Propagation Distances

z	r_1	r_2	r_2/r_1 (expt)	r_2/r_1 (theo)	$\Delta(r_2/r_1)$
0.3	80.5	221.5	2.7516		-0.0438
0.4	92.5	256.5	2.7730		-0.0224
0.5	103.5	290.0	2.8019	2.7954	0.0065
0.6	117.0	323.5	2.7650		-0.0304

generate the optical vortices and acquire the images at $z = 0.6$ m by CCD. Figures 7(a)–7(f) are the intensity patterns of topological charge 1, 3, 5, 10, 15, and 20, respectively. Figure 8(g) shows the normalized transverse intensity profiles.

From the line profiles in Fig. 8(g) and the inner and outer radii, the experimental ratio can be obtained from the pixel number, as shown in Table 5.

According to Table 5, we can see that the experimental data are consistent with numerical values. As the topological charge

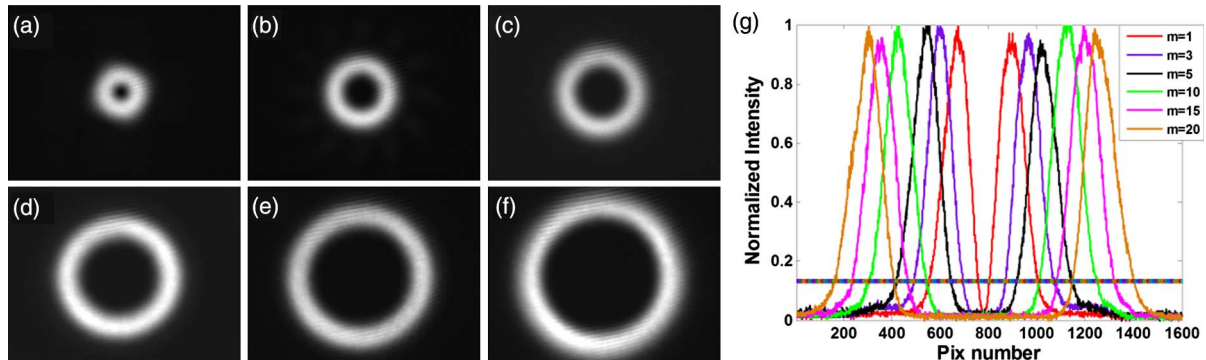


Fig. 8. Generated LG vortex beams with $w_0 = 0.3$ mm and different topological charges: (a) $m = 1$, (b) $m = 3$, (c) $m = 5$, (d) $m = 10$, (e) $m = 15$, (f) $m = 20$ at $z = 0.6$ m, and (g) the normalized transverse intensity profile.

Table 5. Result for the Generated LG Vortex Beams with $w_0 = 0.3$ mm, $z = 0.6$ m, and Different Topological Charges

m	r_1	r_2	r_2/r_1 (expt)	r_2/r_1 (theo)	$\Delta(r_2/r_1)$
1	24.5	226.0	9.2245	9.2663	-0.0418
3	86.5	289.0	3.3410	3.3131	0.0279
5	143.0	351.0	2.4546	2.4953	-0.0407
10	240.5	472.5	1.9647	1.8955	0.0692
15	314.5	538.0	1.7107	1.6824	0.0283
20	382.0	615.5	1.6113	1.5678	0.0435

number increases, the value of r_2/r_1 subsequently decreases and the change rate becomes slower.

5. CONCLUSIONS

We have theoretically shown and demonstrated numerically and experimentally that the inner and outer radii of LG vortex beams with the same topological charge are proportional to the transverse beam size, and the ratio r_2/r_1 depends on the topological charge only and not on the waist radius w_0 and propagation distance z . In addition, we derive the general expressions for the inner and outer radii and show that the ratio decreases gradually as the topological charge increases. These discoveries have very important significance for understanding the dimensional properties of the LG vortex beams, which may be very useful in the fields of optical communication and micromanipulation.

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